CVlytics – AI-Based Resume Analyzer and ATS Scoring System

Project Report

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Program: Bachelor of Science in Artificial Intelligence

Tools & Technologies: Python, Django, Scikit-learn, XGBoost, Pandas, NumPy, NLP, Joblib

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View Demo

1. Project Overview

CVlytics is an AI-powered resume analysis system that predicts the ATS score of a resume using Natural Language Processing and Machine Learning. Users upload a CV in PDF format, which is automatically processed and evaluated to generate a numerical ATS score along with improvement suggestions.

The objective of this system is to provide an automated, fair, and scalable resume evaluation solution for both job seekers and recruiters.

2. System Architecture and Pipeline

Architecture Flow

1. User Uploads Resume (PDF)
2. PDF Text Extraction
3. Text Cleaning & Lemmatization
4. TF-IDF Vectorization (Train-fit only)
5. XGBoost Regression Model
6. ATS Score Prediction
7. Feedback & Error Highlighting
8. Result Display on Web Interface

System Components

Layer	Technology
Frontend	HTML, CSS, JavaScript
Backend	Django
ML Model	XGBoost Regressor
Vectorization	TF-IDF
Database	Supabase / SQLite
File Handling	PyMuPDF / pdfminer

3. Integrated Lab Concepts

Text Preprocessing

Lowercasing, regex cleaning, lemmatization

Feature Engineering

TF-IDF with bigrams

Regression Modeling

ATS score prediction using XGBoost

Hyperparameter Tuning

GridSearchCV

Cross Validation

3-Fold CV

Model Evaluation

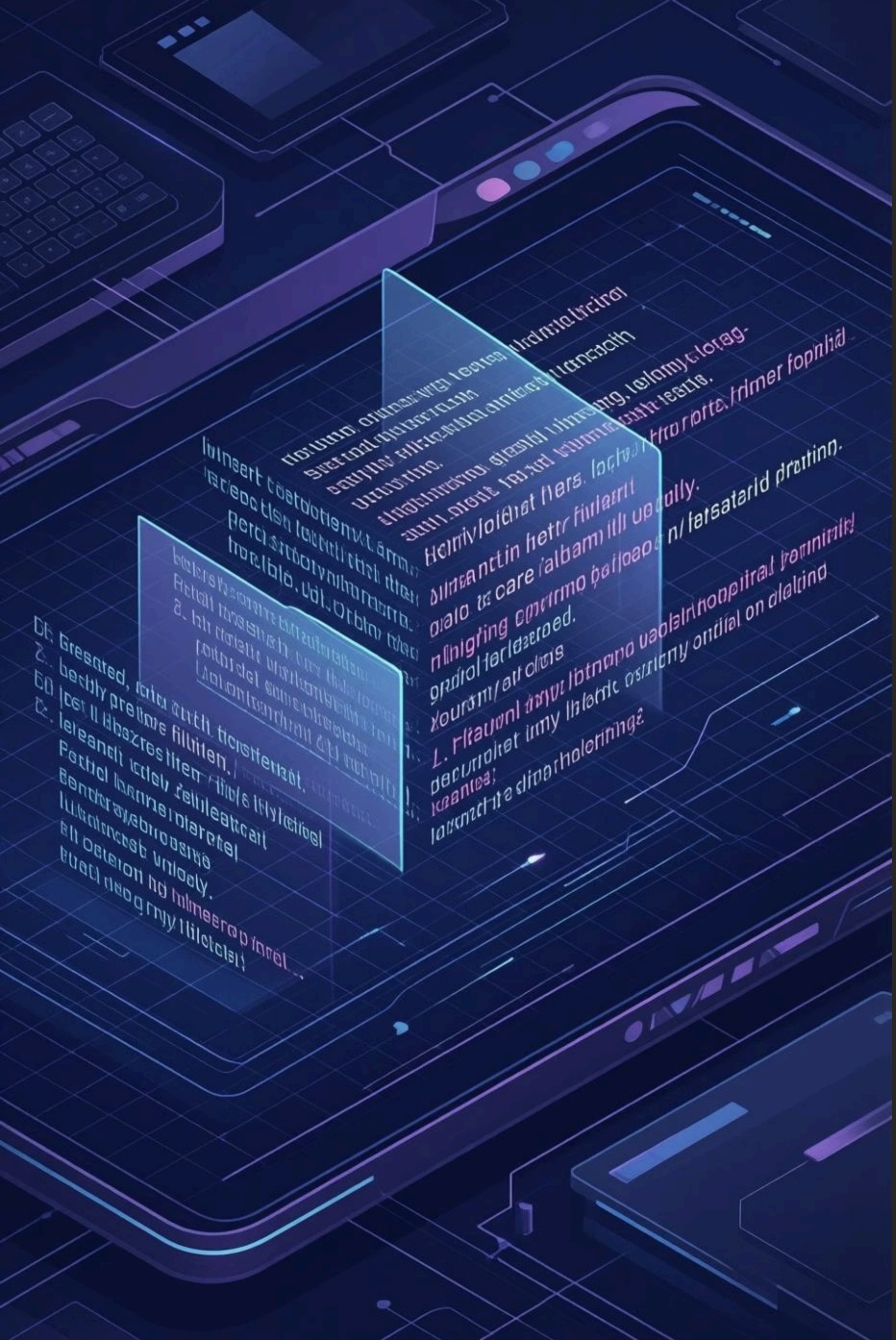
MAE, R^2 Score

Model Serialization

joblib

Deployment

Django model serving



4. Dataset Description and Preprocessing

Dataset

- CSV File containing resume text and ATS scores
- Columns: text, ats_score
- Size: 500+ resumes

Cleaning Process

- Removed rows containing job descriptions instead of resumes
- Converted all text to lowercase
- Removed special characters using regex
- Lemmatized each token using WordNet
- No stopword removal to preserve ATS-relevant keywords

5. Model Design and Training

Feature Extraction

Parameter	Value
Max Features	3000
N-Gram Range	(1,2)
Min DF	3
Max DF	0.9

Model: XGBoost Regressor

Hyperparameter	Value
n_estimators	200, 300
max_depth	4, 6
learning_rate	0.05, 0.1
subsample	0.8
colsample_bytree	0.8

Training Setup

- Train-Test Split: 80% / 20%
- GridSearchCV with 3-fold cross-validation
- Objective: reg:squarederror

Evaluation Metrics

Metric	Result
Mean Absolute Error (MAE)	~5.2
R^2 Score	~0.87

6. Challenges, Limitations and Ethics

Challenges

- ATS scores were manually estimated
- Resume formatting noise
- PDF parsing inconsistencies

Limitations

- Model only predicts generic ATS score
- No job-specific resume optimization

Ethics & Fairness

- No demographic features used
- Scoring purely based on resume content
- No personal data stored after processing

7. Setup and Execution Instructions

Environment Setup

```
conda create -n cvlytics python=3.10
conda activate cvlytics
pip install -r requirements.txt
```

Run Django Server

```
python manage.py makemigrations
python manage.py migrate
python manage.py runserver
```

Hardware Requirements

Component	Specification
RAM	8 GB
CPU	Intel i5 or equivalent
OS	Windows / Linux

8. Sample API Command

POST /analyze_cv/

Input: resume.pdf

Output: Predicted ATS Score + Resume Feedback

9. Conclusion

CVlytics is a complete AI-driven ATS scoring platform that uses TF-IDF and XGBoost regression to predict resume quality. The system demonstrates how real-world machine learning models can be trained, evaluated, deployed, and served within a full-stack Django application.